

Question 1: Collection Creation

```
switched to db sep
sep> db.createCollection("emp")
| db.createCollection("dept")
{ ok: 1 }
sep> |
```

Question 2: Inserting Documents

```
$project: {
  _id: 0,
  ename: 1,
  dname: "$department.dname",
  income: {
    $add: [
      "$sal",
      { $ifNull: ["$comm", 0] }
    ]
  }
},
{
  $sort: {
    income: 1
  }
}
])
[
{ ename: 'SMITH', dname: 'RESEARCH', income: 800 },
{ ename: 'JAMES', dname: 'SALES', income: 950 },
{ ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
{ ename: 'MARTIN', dname: 'ACCOUNTING', income: 1250 }
]
```

```

sep> db.emp.insertMany([
  { _id: 7369, ename: "SMITH", job: "CLERK", mgr: 7902, hiredate: "1980-12-17", sal: 800, comm: 0, deptno: 20 },
  { _id: 7499, ename: "ALLEN", job: "SALESMAN", mgr: 7698, hiredate: "1981-02-20", sal: 1600, comm: 300, deptno: 30 },
  { _id: 7521, ename: "WARD", job: "SALESMAN", mgr: 7698, hiredate: "1981-02-22", sal: 1250, comm: 500, deptno: 30 },
  { _id: 7566, ename: "JONES", job: "MANAGER", mgr: 7839, hiredate: "1981-04-02", sal: 2975, comm: 0, deptno: 20 },
  { _id: 7698, ename: "BLAKE", job: "MANAGER", mgr: 7839, hiredate: "1981-05-01", sal: 2850, comm: 0, deptno: 30 },
  { _id: 7782, ename: "CLARK", job: "MANAGER", mgr: 7839, hiredate: "1981-06-09", sal: 2450, comm: 0, deptno: 10 },
  { _id: 7788, ename: "SCOTT", job: "ANALYST", mgr: 7566, hiredate: "1987-04-19", sal: 3000, comm: 0, deptno: 20 },
  { _id: 7839, ename: "KING", job: "PRESIDENT", mgr: null, hiredate: "1981-11-17", sal: 5000, comm: 0, deptno: 10 },
  { _id: 7844, ename: "TURNER", job: "SALESMAN", mgr: 7698, hiredate: "1981-09-08", sal: 1500, comm: 0, deptno: 30 },
  { _id: 7876, ename: "ADAMS", job: "CLERK", mgr: 7788, hiredate: "1987-05-23", sal: 1100, comm: 0, deptno: 20 },
  { _id: 7900, ename: "JAMES", job: "CLERK", mgr: 7698, hiredate: "1981-12-03", sal: 950, comm: 0, deptno: 30 },
  { _id: 7902, ename: "FORD", job: "ANALYST", mgr: 7566, hiredate: "1981-12-03", sal: 3000, comm: 0, deptno: 20 },
  { _id: 7934, ename: "MILLER", job: "CLERK", mgr: 7782, hiredate: "1982-01-23", sal: 1300, comm: 0, deptno: 10 }
])
{
  acknowledged: true,
  insertedIds: {
    '0': 7369,
    '1': 7499,
    '2': 7521,
    '3': 7566,
    '4': 7698,
    '5': 7782,
    '6': 7788,
    '7': 7839,
    '8': 7844,
    '9': 7876,
    '10': 7900,
    '11': 7902,
    '12': 7934
  }
}
sep>

```

Question 3: Updating Documents

```

    $project: {
      _id: 0,
      ename: 1,
      dname: "$department.dname",
      income: {
        $add: [
          "$sal",
          { $ifNull: ["$comm", 0] }
        ]
      }
    },
    $sort: {
      income: 1
    }
  ]
})
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MILLER', dname: 'RESEARCH', income: 1300 },
  { ename: 'TURNER', dname: 'SALES', income: 1500 },
  { ename: 'SCOTT', dname: 'RESEARCH', income: 3000 },
  { ename: 'FORD', dname: 'RESEARCH', income: 3000 },
  { ename: 'JONES', dname: 'RESEARCH', income: 2975 },
  { ename: 'WARD', dname: 'SALES', income: 1250 },
  { ename: 'ALLEN', dname: 'SALES', income: 1600 },
  { ename: 'BLAKE', dname: 'RESEARCH', income: 2850 },
  { ename: 'CLARK', dname: 'RESEARCH', income: 2450 },
  { ename: 'KING', dname: 'RESEARCH', income: 5000 }
]

```

```

}
sep> db.emp.updateMany(
|   { deptno: 10 },
|   { $inc: { sal: 100 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
sep> db.emp.find({ deptno: 10 })
[
  {
    _id: 7782,
    ename: 'CLARK',
    job: 'MANAGER',
    mgr: 7839,
    hiredate: '1981-06-09',
    sal: 2550,
    comm: 0,
    deptno: 10
  },
  {
    _id: 7839,
    ename: 'KING',
    job: 'PRESIDENT',
    $project: {
      _id: 0,
      ename: 1,
      dname: "$department.dname",
      income: {
        $add: [
          "$sal",
          { $ifNull: ["$comm", 0] }
        ]
      }
    }
  },
  {
    $sort: {
      income: 1
    }
  }
]
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MARTIN', dname: 'SALES', income: 1250 },
  { ename: 'WATSON', dname: 'RESEARCH', income: 1300 },
  { ename: 'SCOTT', dname: 'OPERATIONS', income: 1450 },
  { ename: 'COOPER', dname: 'OPERATIONS', income: 1550 },
  { ename: 'JONES', dname: 'OPERATIONS', income: 1600 },
  { ename: 'FORD', dname: 'OPERATIONS', income: 1700 },
  { ename: 'BLAKE', dname: 'OPERATIONS', income: 1800 },
  { ename: 'MILLER', dname: 'OPERATIONS', income: 1900 },
  { ename: 'TURNER', dname: 'OPERATIONS', income: 2000 },
  { ename: 'SIMPSON', dname: 'OPERATIONS', income: 2100 },
  { ename: 'PERCIVAL', dname: 'OPERATIONS', income: 2200 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 2300 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 2400 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 2500 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 2600 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 2700 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 2800 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 2900 },
  { ename: 'MCCOY', dname: 'OPERATIONS', income: 3000 }
]

```

Question 4: Querying with Filtering and Sorting

```
]
sep> db.emp.find(
|   { sal: { $gt: 1000 } },
|   { _id: 0, ename: 1, job: 1, deptno: 1, sal: 1 }
| ).sort({
|   deptno: 1,
|   job: 1,
|   sal: -1
| })
[
  { ename: 'MILLER', job: 'CLERK', sal: 1400, deptno: 10 },
  { ename: 'CLARK', job: 'MANAGER', sal: 2550, deptno: 10 },
  { ename: 'KING', job: 'PRESIDENT', sal: 5100, deptno: 10 },
  { ename: 'SCOTT', job: 'ANALYST', sal: 3000, deptno: 20 },
  { ename: 'FORD', job: 'ANALYST', sal: 3000, deptno: 20 },
  { ename: 'ADAMS', job: 'CLERK', sal: 1100, deptno: 20 },
  { ename: 'JONES', job: 'MANAGER', sal: 2975, deptno: 20 },
  { ename: 'BLAKE', job: 'MANAGER', sal: 2850, deptno: 30 },
  { ename: 'ALLEN', job: 'SALESMAN', sal: 1600, deptno: 30 },
  { ename: 'TURNER', job: 'SALESMAN', sal: 1500, deptno: 30 },
  { ename: 'WARD', job: 'SALESMAN', sal: 1250, deptno: 30 }
]
sep> |
```

Question 5: Deleting Documents

```
    $project: {
      _id: 0,
      ename: 1,
      dname: "$department.dname",
      income: {
        $add: [
          "$sal",
          { $ifNull: ["$comm", 0] }
        ]
      }
    },
  },
  {
    $sort: {
      income: 1
    }
  }
])
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MILLER', dname: 'SALES', income: 1400 },
  { ename: 'SCOTT', dname: 'RESEARCH', income: 3000 },
  { ename: 'FORD', dname: 'RESEARCH', income: 3000 },
  { ename: 'JONES', dname: 'RESEARCH', income: 2975 },
  { ename: 'BLAKE', dname: 'RESEARCH', income: 2850 },
  { ename: 'TURNER', dname: 'RESEARCH', income: 1500 },
  { ename: 'WARD', dname: 'RESEARCH', income: 1250 }
]
```

```

]
sep> db.dept.deleteOne({ _id: 40 })
{ acknowledged: true, deletedCount: 1 }
sep> db.dept.find()
[
  { _id: 10, dname: 'ACCOUNTING', loc: 'NEW YORK' },
  { _id: 20, dname: 'RESEARCH', loc: 'DALLAS' },
  { _id: 30, dname: 'SALES', loc: 'CHICAGO' }
]
sep> |

```

Part 2: Grouping and Aggregation Pipeline Operations

Question 1: Aggregation with Grouping, Matching, and Projection

```

    $project: {
      _id: 0,
      ename: 1,
      dname: "$department.dname",
      income: {
        $add: [
          "$sal",
          { $ifNull: ["$comm", 0] }
        ]
      }
    },
    {
      $sort: {
        income: 1
      }
    }
  ]
})
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MARTIN', dname: 'ACCOUNTING', income: 1250 }
]

```

```

sep> db.emp.aggregate([
  {
    $match: {
      job: { $ne: "PRESIDENT" }
    }
  },
  {
    $group: {
      _id: "$job",
      total_sal: { $sum: "$sal" },
      max_sal: { $max: "$sal" },
      min_sal: { $min: "$sal" },
      avg_sal: { $avg: "$sal" }
    }
  },
  {
    $project: {
      _id: 0,
      job: "$_id",
      total_sal: 1,
      max_sal: 1,
      min_sal: 1,
      avg_sal: 1
    }
  }
])

[
  {
    $project: {
      _id: 0,
      ename: 1,
      dname: "$department.dname",
      income: {
        $add: [
          "$sal",
          { $ifNull: ["$comm", 0] }
        ]
      }
    }
  },
  {
    $sort: {
      income: 1
    }
  }
])

[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MARTIN', dname: 'ACCOUNTING', income: 1250 },
  { ename: 'WATSON', dname: 'RESEARCH', income: 1300 },
  { ename: 'SCOTT', dname: 'OPERATIONS', income: 1450 },
  { ename: 'TUCKER', dname: 'OPERATIONS', income: 1550 },
  { ename: 'DEVERAUX', dname: 'OPERATIONS', income: 1600 },
  { ename: 'MILLER', dname: 'OPERATIONS', income: 1700 },
  { ename: 'FORD', dname: 'OPERATIONS', income: 3100 },
  { ename: 'JONES', dname: 'OPERATIONS', income: 3200 },
  { ename: 'BLAKE', dname: 'OPERATIONS', income: 3300 },
  { ename: 'ALLEN', dname: 'OPERATIONS', income: 3400 },
  { ename: 'WARD', dname: 'OPERATIONS', income: 3500 },
  { ename: 'MARTIN', dname: 'OPERATIONS', income: 3600 },
  { ename: 'SMITH', dname: 'OPERATIONS', income: 3700 },
  { ename: 'JONES', dname: 'OPERATIONS', income: 3800 },
  { ename: 'BLAKE', dname: 'OPERATIONS', income: 3900 },
  { ename: 'ALLEN', dname: 'OPERATIONS', income: 4000 },
  { ename: 'WARD', dname: 'OPERATIONS', income: 4100 },
  { ename: 'MARTIN', dname: 'OPERATIONS', income: 4200 },
  { ename: 'SMITH', dname: 'OPERATIONS', income: 4300 },
  { ename: 'JONES', dname: 'OPERATIONS', income: 4400 },
  { ename: 'BLAKE', dname: 'OPERATIONS', income: 4500 },
  { ename: 'ALLEN', dname: 'OPERATIONS', income: 4600 },
  { ename: 'WARD', dname: 'OPERATIONS', income: 4700 },
  { ename: 'MARTIN', dname: 'OPERATIONS', income: 4800 },
  { ename: 'SMITH', dname: 'OPERATIONS', income: 4900 },
  { ename: 'JONES', dname: 'OPERATIONS', income: 5000 }
]

```

Question 2: Aggregation Pipeline Output

```
$project: {
  _id: 0,
  ename: 1,
  dname: "$department.dname",
  income: {
    $add: [
      "$sal",
      { $ifNull: ["$comm", 0] }
    ]
  }
}
},
{
  $sort: {
    income: 1
  }
}
])
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MARTIN', dname: 'ACCOUNTING', income: 1250 }
]
```

```
{
  total_sal: 4350,
  max_sal: 1600,
  min_sal: 1250,
  avg_sal: 1450,
  job: 'SALESMAN'
},
{
  total_sal: 4250,
  max_sal: 1400,
  min_sal: 800,
  avg_sal: 1062.5,
  job: 'CLERK'
},
{
  total_sal: 8375,
  max_sal: 2975,
  min_sal: 2550,
  avg_sal: 2791.6666666666665,
  job: 'MANAGER'
},
{
  total_sal: 6000,
  max_sal: 3000,
  min_sal: 3000,
  avg_sal: 3000,
  job: 'ANALYST'
}

  $project: {
    _id: 0,
    ename: 1,
    dname: "$department.dname",
    income: {
      $add: [
        "$sal",
        { $ifNull: ["$comm", 0] }
      ]
    }
  }
},
{
  $sort: {
    income: 1
  }
}
])
```

Question 3: Lookup, Unwind, and Calculated Fields

```
$project: {
  _id: 0,
  ename: 1,
  dname: "$department.dname",
  income: {
    $add: [
      "$sal",
      { $ifNull: ["$comm", 0] }
    ]
  }
},
},
{
  $sort: {
    income: 1
  }
}
])
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MARTIN', dname: 'SALES', income: 1250 },
  { ename: 'BLAKE', dname: 'RESEARCH', income: 1050 },
  { ename: 'CLARK', dname: 'RESEARCH', income: 800 },
  { ename: 'MILLER', dname: 'SALES', income: 750 },
  { ename: 'SCOTT', dname: 'RESEARCH', income: 950 },
  { ename: 'TURNER', dname: 'SALES', income: 800 },
  { ename: 'WARD', dname: 'SALES', income: 900 },
  { ename: 'ZELNY', dname: 'RESEARCH', income: 1000 }
]
```

```

]
sep> db.emp.aggregate([
  {
    $lookup: {
      from: "dept",
      localField: "deptno",
      foreignField: "_id",
      as: "department"
    }
  },
  {
    $unwind: "$department"
  },
  {
    $project: {
      _id: 0,
      ename: 1,
      dname: "$department.dname",
      income: {
        $add: [
          "$sal",
          { $ifNull: ["$comm", 0] }
        ]
      }
    }
  },
  {
    $sort: {
      $project: {
        _id: 0,
        ename: 1,
        dname: "$department.dname",
        income: {
          $add: [
            "$sal",
            { $ifNull: ["$comm", 0] }
          ]
        }
      }
    }
  },
  {
    $sort: {
      income: 1
    }
  }
])
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MARTIN', dname: 'SALES', income: 1250 },
  { ename: 'BLAKE', dname: 'MANAGEMENT', income: 1300 },
  { ename: 'CLARK', dname: 'MANAGEMENT', income: 1400 },
  { ename: 'SCOTT', dname: 'MANAGEMENT', income: 1500 },
  { ename: 'KING', dname: 'MANAGEMENT', income: 1600 }
]

```

```

    $project: {
      _id: 0,
      ename: 1,
      dname: "$department.dname",
      income: {
        $add: [
          "$sal",
          { $ifNull: ["$comm", 0] }
        ]
      }
    },
  },
  {
    $sort: {
      income: 1
    }
  }
])
[
  { ename: 'SMITH', dname: 'RESEARCH', income: 800 },
  { ename: 'JAMES', dname: 'SALES', income: 950 },
  { ename: 'ADAMS', dname: 'RESEARCH', income: 1100 },
  { ename: 'MARTIN', dname: 'ACCOUNTING', income: 1250 }
]

```